

Sathyaruba M

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PROFESSIONAL SUMMARY

Pre-final year B.E. AI & Data Science student at Velammal Engineering College (CGPA 8.58) with strong expertise in Computer Vision, Machine Learning, and Big Data. Multiple hackathon winner, Presidio internship selectee, and builder of startup-level AI systems targeting real-world Indian problems. Seeking AI/ML internship or full-time roles.

EDUCATION

B.E. – Artificial Intelligence & Data Science

2023 – 2027

Velammal Engineering College, Chennai | CGPA: **8.58 / 10**

Higher Secondary (Class XII)

2022 – 2023

Kamaraj Matriculation Higher Secondary School, Kovilpatti | 82%

TECHNICAL SKILLS

Languages

Python, C, C++, SQL

Frameworks

TensorFlow, PyTorch, Keras, Scikit-learn, PySpark

Libraries

NumPy, Pandas, Matplotlib, Seaborn, OpenCV

Tools

Firebase, Docker, Git

Data / EDA

Data Cleaning, Feature Engineering, Outlier Detection, Statistical Analysis, Data Visualization

Domains

Computer Vision, Machine Learning, Deep Learning, ADAS

PROJECTS

AI360Drive – Advanced ADAS for Indian Roads

Startup Concept | Tech: YOLOv8n, OpenCV, Python, Radar Fusion, MediaPipe

GitHub: https://github.com/Sathya-2006/SMART_AI

- Designed Level 3+ ADAS using **YOLOv8n** — achieved **92% detection accuracy at 25 FPS** trained on 15k+ annotated Indian road images (vehicles, pedestrians, animals, potholes).
- Built real-time risk prediction engine fusing camera + radar + historical data — generates accident-probability alerts within **200ms latency**, reducing false positives by **~30%**.
- Implemented behavior-based pedestrian crossing prediction, pothole detection, and driver drowsiness monitoring — camera-only setup, no LiDAR required.

Intruder Detection Smart Alert System for Visually Impaired

AI/CV Project | Tech: YOLOv8s, MediaPipe, Firebase, Twilio, Python | 2nd Prize – Mini Project Expo

GitHub: github.com/Sathya-2006/Intruder-Detection

- Built real-time security system using **YOLOv8s** trained on 8k+ images — achieves **89% weapon/mask detection accuracy at 20 FPS** with sub-500ms threat classification.
- Integrated **MediaPipe** pose estimation for behavior analysis + **Twilio** automated emergency SMS with GPS tagging — cutting manual alert delay by **~80%**.

ShelfSense AI – Retail Shelf Intelligence System

Startup Concept | Tech: YOLOv8, OpenCV, Flask, Firebase, Python

GitHub: github.com/Sathya-2006/ShelfSense-AI

- Designed AI shelf monitoring using **YOLOv8 + OpenCV** on existing CCTV feeds — detects out-of-stock gaps and misplacements with **zero additional hardware cost**.
- Firebase real-time alerts to staff, reducing estimated manual stock-audit time by **70%**; scalable architecture supports multi-store deployment.

Machine Failure Prediction Using PySpark

BigData / ML | Tech: PySpark, RandomForest, Python, Pandas | 1st Prize – Course Project

GitHub: https://github.com/Sathya-2006/machinefailure_prediction

- Built end-to-end PySpark pipeline on **50k+ sensor records** — outlier handling, feature engineering, **Random Forest classifier** for predictive maintenance with real-time input support.
- Ranked **1st out of 15 teams**; model achieved strong precision/recall balance validated via cross-validation on held-out test split.

Heart Disease Prediction

ML Project | Tech: LogisticRegression, Scikit-learn, Python

GitHub: https://github.com/Sathya-2006/Heartdisease_prediction

- Built **logistic regression** model on Cleveland Heart Disease dataset (303 records) — full pipeline: imputation, encoding, scaling; evaluated with **85%+ accuracy** and confusion matrix.

Iris Classification Pipeline

ML Project | Tech: Scikit-learn, Python, Decision Tree, SVM, Random Forest

GitHub: github.com/Sathya-2006/Iris-Classification

- Multi-algorithm pipeline with cross-validation and **hyperparameter tuning** — Voting Classifier achieved best accuracy; demonstrated ensemble learning advantage over single models.

ACHIEVEMENTS

- **Presidio Internship – Dept. Selection:** Selected among **14 students** company-wide for outstanding technical & academic performance.
- **2nd Prize** – SindhanAI'25 Hackathon, SRM TRP College of Engineering, Trichy.
- **2nd Prize** – Velammal Hackathon, Velammal Engineering College.
- **2nd Prize** – Mini Project Expo, Velammal Engineering College (Intruder Detection System).
- **1st Prize** – PySpark Course Project, Velammal Engineering College (out of 15 teams).
- **2nd Prize** – Web Designing Competition, Jeppiar Institute of Technology.
- **SIH Contributor** – Software implementation in team recognized at inter-college level.

LEADERSHIP

- Led **2-member team** at Mini Project Expo — presented Intruder Detection System, secured **2nd place** at Velammal Engineering College.
- Led **4-member team** in PySpark course project — implemented feature selection pipeline, **won 1st place** out of 15 competing teams.
- Led **2-member team** as visually impaired team leader — achieved **2nd place out of 400 participants** for innovative AI solution.
- **MSME Innovation Project** – Idea selected at inter-college level; led **5-member team** in design and presentation out of 40 teams.

SOFT SKILLS

Leadership · Teamwork · Creative Thinking · Project Management · Problem Solving